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Is periapical surgery follow-up with only two-dimensional radiographs reliable? A retrospective cohort type sensitivity study

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Introduction

The radiographic changes in persistent apical periodontitis are a characteristic sign of this disorder (1), and it is very important to know the extent of the periapical bone defects before endodontic microsurgery to facilitate treatment and improve the prognosis of the tooth (2). Such changes range from the widening of the periodontal ligament to a well-defined periapical area (3).

At present, periapical radiography is the most common technique used in application to persistent apical periodontitis, both for the initial diagnosis and for follow-up (4,5). Panoramic radiography is not as frequently used as periapical radiography, though it does offer a comprehensive and rapid two-dimensional (2D) view of the jaws, making it easy to use in daily clinical practice (6). The main drawbacks of two-dimensional diagnostic methods in their reliability rely in its limited information about size, extension and location of the periapical lesion (7), because of compression of three-dimensional structures, geometric distortion and anatomic noise obscuring diagnostic clarity of the region of interest (8). Moreover, in the vestibular plane of periapical radiographs, the information provided is limited due to bone superimposition that makes it difficult to observe periapical radiolucent areas (4). In addition, the size of the periapical radiolucency may be affected by the orientation of the film and tube head (9).

Some authors associate the size of the periapical lesion to the prognosis of endodontic surgery, reporting better results with smaller areas (10,11), while other investigators have found no relationship between previous size and the prognosis of the tooth (12,13). In any case, prior to endodontic microsurgery, it is important to precisely establish the extent of the lesion in order to optimize the surgical procedure (2). During the follow-up, it is also important to assess the size of the periapical area in order to adequately assess the outcome of endodontic microsurgery, especially when the tooth needs to be used for prosthetic purposes.

Cone-beam computed tomography (CBCT) has proven its superiority over two-dimensional radiographs (avoiding anatomical noise, compression of three-dimensional anatomy and geometric distortion, and allowing better assessment of the root canal anatomy) (14,15). Besides, CBCT has been recommended in cases where apical surgery is being considered (16). No studies to date have investigated the sensitivity of the mentioned two-

dimensional radiographic techniques (periapical and panoramic) versus CBCT both before surgery and in the course of follow-up. Likewise, no comparisons have been made of the periapical areas obtained with these two-dimensional techniques versus the volumes obtained with CBCT before and after surgical treatment.

The present study was carried out to analyze the sensitivity of two-dimensional radiographic techniques and their reliability in evaluating the size of periapical bone defects, both preoperatively and during the follow-up period.

Material and Methods

- Study design and setting

The present retrospective cohort type sensitivity study is reported according to the STARD statement (Standards for Reporting of Diagnostic Accuracy Studies) (17).

- Patient selection

A consecutive cohort of 298 patients underwent endodontic microsurgery between January 2015 and January 2018 at the Stomatology Department of the University of Valencia (Valencia, Spain). The cohort was evaluated and screened for study eligibility by the first author (A.R.A.) before study entry.

Inclusion criteria: Patients with a single tooth subjected to primary endodontic microsurgery, involving the use of three radiological diagnostic methods (CBCT, periapical radiography and panoramic radiography) affording images of sufficient quality to allow linear measurements of the periapical lesions, and who reported for follow-up examinations after 6-48 months postsurgery.

Exclusion criteria: The adoption of regenerative procedures (bone grafts, membranes).

- Endodontic microsurgery

Modern microsurgical techniques were performed in all the patients (18). High magnification was used with a rigid endoscope (Karl Storz-Endoskope, Tuttlingen, Germany), with ultrasonic preparation (Piezon® Master 400, EMS®, Electro Medical Systems S.A., Switzerland) and root-end fillings with mineral trioxide aggregate (MTA) (ProRoot MTA White, Dentsply Tulsa Dental, Tulsa, OK, USA). All surgeries were carried out by the same oral surgeon (M.P.D.).

- Imaging techniques

Index test: Periapical and panoramic radiographs: The periapical radiography images were obtained with a Gendex expert D unit (Gendex Dental Systems, Hat-

field, USA). A phosphor-plate system was used (Vista Scan®, Dürr Dental AG, Bietigheim-Bissingen, Germany). Images were taken according to the parallel technique using a film holder (Super-Bite®, Kerr, Bioggio, Switzerland). The exposure parameters were: 65 kV, 7 mA and 0.08~0.12 sec. The distance between the sensor and X-ray tube was 30 cm. Images were processed and stored by the processor DBSWIN (Dürr Dental, Germany).

The panoramic radiographs were obtained with a Planmeca ProMax 3D Classic CBCT unit (Planmeca, Helsinki, Finland), and the Planmeca Romexis® application was used for processing the images (version 4.5.2; Planmeca). The exposure parameters were: 68 kV, 10 mA and 19 sec.

Reference standard: CBCT: The CBCT images were obtained with a Planmeca ProMax 3D Classic CBCT unit (Planmeca, Helsinki, Finland), and the Planmeca Romexis® Viewer application was used for processing the images. A limited field of view (FOV) (40 x 40 mm) with 12 sec duration at 90 kV, 8 mA and voxel size 0.15 mm was used for CBCT imaging.

- Test sensitivity

For both the two-dimensional radiographic techniques and CBCT, a periapical lesion was defined as a periapical radiolucency in contact with the apical part of the root that exceeded at least twice the width of the periodontal ligament space. Also, in CBCT images, the lesion had to be visible in at least two image planes (16). The sensitivity of the two-dimensional techniques (index test) compared with CBCT (reference standard) was assessed according to the imaging stage involved (preoperative and follow-up). All preoperative and follow-up images were evaluated independently by the first author (A.R.A) and a blinded assessor (D.P.O.). Disagreements were resolved by discussion with a third advisor (D.S.P.). The two-dimensional radiographic im-

ages were evaluated first, followed by the CBCT images at least two weeks later.

Examiners were calibrated, and the appraisals were conducted under standardized conditions.

- Lesion area and volume calculation

The radiographic images were evaluated jointly by the same two examiners (A.R.A. and D.P.O.). During the measurements, the examiners discussed and reached a consensus regarding the limits of the areas. Previously, to ensure calibration of measures and inter-examiner reliability, 10 images per radiographic technique from patients not involved in the present study and presenting periapical areas were evaluated twice, with a one-week interval.

Periapical and panoramic radiographs assessment: The area measurements were done using the Itk-Snap program (free access software, <http://www.itksnap.org>). Two investigators (A.R.A and D.P.O) used this program to measure the images preoperatively and follow-up in square millimeters. The examiners made manual tracing. The freehand selection was used to trace out the lesion contour (after calibrating the software's scale according to the size of the active sensor area). Then, the area value was measured and recorded as previously reported (19). If a multiple root tooth had more than one periapical lesion, the individual defects were calculated and then added together to obtain a total defect area (20) (Fig. 1, Fig. 2).

Tomographic imaging assessment: For the volume measurements in the CBCT images, the dedicated Planmeca Romexis® application (version 4.5.2; Planmeca) was used. This software allowed volume calculation of the periapical bone defects through manual tracing of the border of the lesion across the different slices (axial, coronal and sagittal), which contained the bone defects and, finally, the program calculated the total volume in cubic millimeters (Fig. 3).

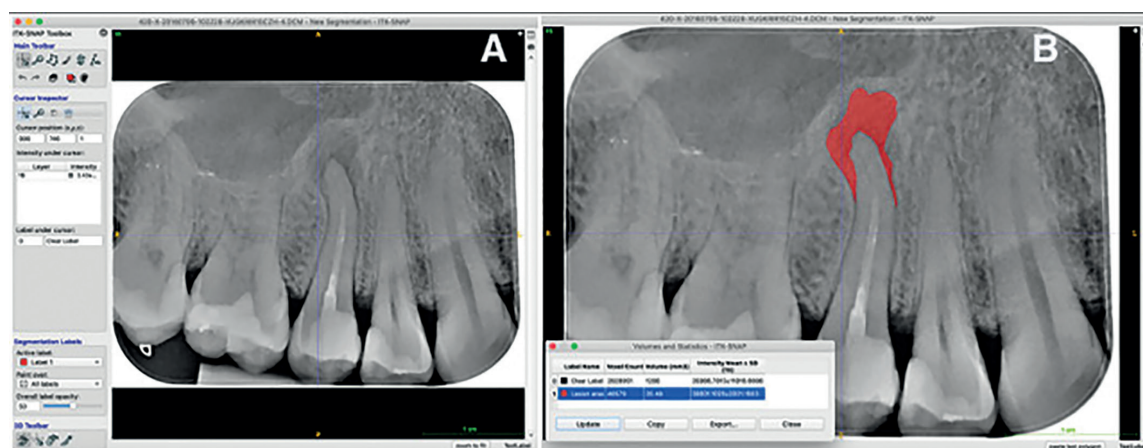


Fig. 1: a. Periapical radiography showing a periapical area in a 2nd upper right premolar before endodontic microsurgery. b. Measurement performed manually in the program Itk-Snap (20.49 mm²).

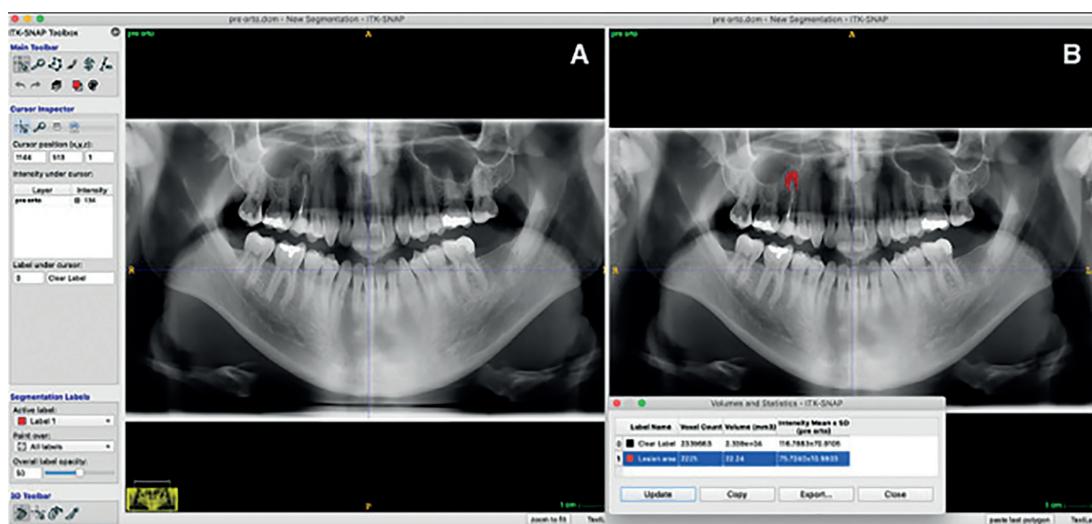


Fig. 2: a. Panoramic radiography showing the periapical area in the same 2nd premolar. b. Measurement performed manually in the program Itk-Snap (22.24 mm²).

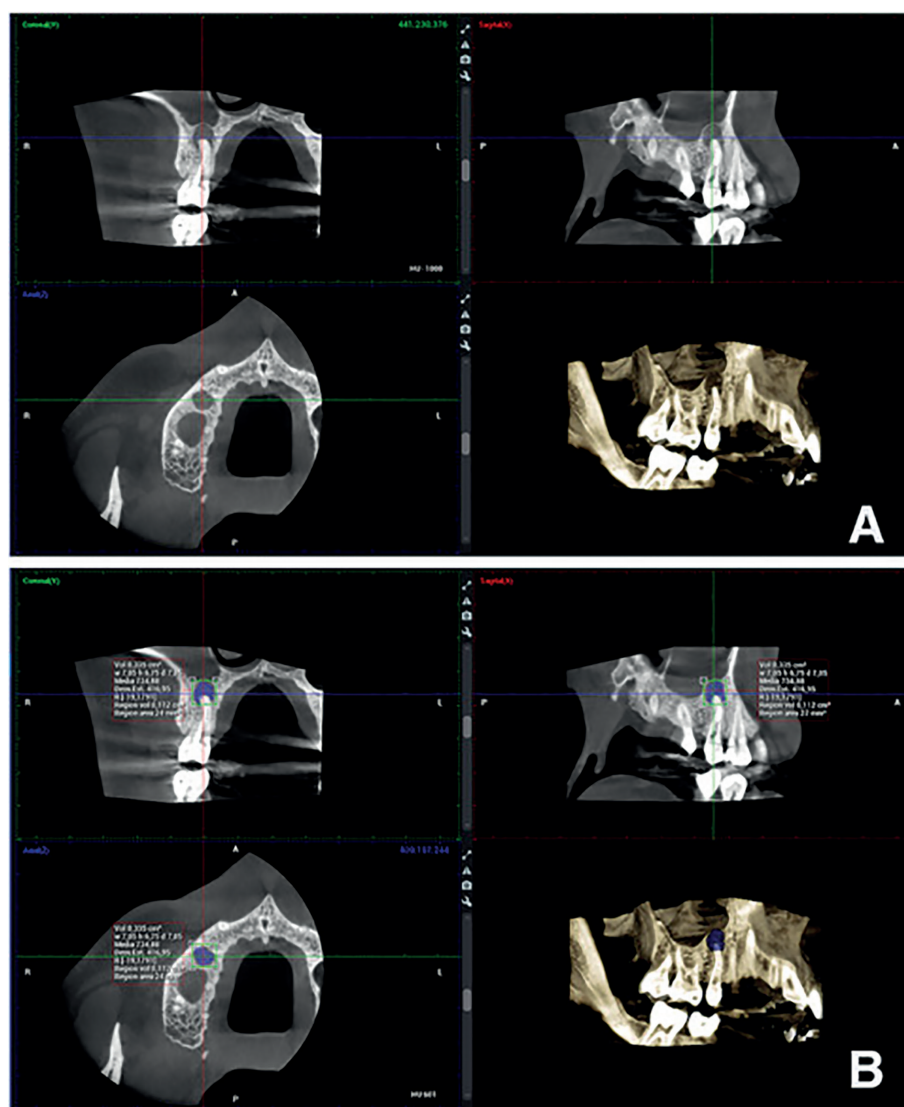


Fig. 3: a. CBCT images of the same 2nd premolar with sagittal, axial and coronal sections. b. Volume measurements performed in the Planmeca Software (112 mm³).

Standardization assessment: All images were evaluated on a 21.5-inch monitor (iMac; Apple, Cupertino, CA, USA) with a screen resolution of 4096 x 2304 pixels and located in a quiet room with subdued lighting.

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Conflict of interest

The authors declare that they have no conflicts of interest concerning this study.

Ethics

The project is approved by the ethical committee of the University of Valencia (approval number H1523379927800).

Authors contributions

A. Ramis-Alario, contributed to conception and design, drafted and critically revised the manuscript; D. Soto-Peñaloza, contributed to conception, and analysis, drafted and critically revised the manuscript; B. Tarazona, contributed to design, drafted and critically revised the manuscript; M. Peñarrocha-Diogo, contributed to acquisition, analysis, or interpretation and critically revised the manuscript; D. Peñarrocha-Oltra contributed to data interpretation and critically revised the manuscript. All authors gave final approval and agreed to be accountable for all aspects of the work.